2o2dy

File

Name: K04-EV-stage1
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1 Overview

Calculation method		ISO 6336:2019	
Drawing or article number:			
Gear 1:		z1(z1z2)	
Gear 2:		z2(z1z2)	
		------- Gear 1 ----------	Gear 2 -------
Power (kW)	[P]	234.572	
Speed (1/min)	[n]	7000.0	2983.6
Number of load cycles (in mio.)	[NL]	3990.000	1700.656
Torque (Nm)	[T]	320.0	750.8
Application factor	[KA]	1.00	
Required service life (h)	[H]	9500.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:		Right flank	
Gear 1 direction of rotation:		Clockwise	

2 Tooth geometry

Geometry calculation according to		ISO 21771:2007 (withdrawn)	
		------- Gear 1 ----------	Gear 2 -------
Center distance (mm)	[a]	108.000	
Center distance tolerance		ISO 286:2010 Measure js7	
Normal module (mm)	[mn]	2.1500	
Transverse module (mm)	[mt]	2.4582	
Normal diametral pitch (1/in)	[Pnd]	11.81395	
Transverse diametral pitch (1/in)	[Ptd]	10.33272	
Normal pressure angle (°)	[αn]	18.0000	
Helix angle at reference circle (°)	[β]	29.0000	
Number of teeth	[z]	26	61
Facewidth (mm)	[b]	31.00	29.00
Hand of gear		left	right
Accuracy grade	[Q-ISO 1328:2013]	A6	A6
Inner diameter (mm)	[di]	25.00	128.54
Shaft diameter (mm)	[ds]	25.00	128.54
Inner diameter of gear rim (mm)	[dbi]	0.00	0.00
		------- Gear 1 ----------	Gear 2 -------

3 Materials

Gear 1

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

Gear 2

18CrNiMo7-6, Case hardening steel, case-hardened, ISO 6336-5 Figure 9/10 (MQ), Core hardness ≥25HRC Jominy J=12mm<HRC28

------- Gear 1 ---------- Gear 2 -------

Surface hardness		HRC 61	HRC 61
Material treatment according to ISO 6336:2006 Normal, Life factors Z_{NT} and Y_{NT} ≥ 0.85			
Infinite life strength for tooth root stress (N/mm²)	[σFlim]	425.00	425.00
Infinite life strength for Hertzian pressure (N/mm²)	[σHlim]	1500.00	1500.00
Young's modulus (N/mm²)	[E]	206000	206000
Poisson's ratio	[v]	0.300	0.300
Tensile strength (N/mm²)	[σB]	1200.00	1200.00
Yield point (N/mm²)	[σS]	850.00	850.00

3.1 Gear roughness

		------- Gear 1 ----------	Gear 2 -------
Arithmetic mean roughness value R_a, flank (µm)	[RAH]	0.60	0.60
Arithmetic mean roughness value R_a, root (µm)	[RAF]	3.00	3.00
Mean peak-to-valley roughness R_z, flank (µm)	[RZH]	4.80	4.80
Mean peak-to-valley roughness R_z, root (µm)	[RZF]	20.00	20.00

3.2 Lubrication

Lubrication type	Oil bath lubrication	
Type of oil	ISO-VG 46	
Lubricant base	Mineral-oil base	
Lubricant factor	[XL]	1.000
Oil nominal kinematic viscosity at 40°C (mm²/s)	[v40]	46.00
Oil nominal kinematic viscosity at 100°C (mm²/s)	[v100]	6.80
Specific density at 15°C (kg/dm³)	[ρ]	0.878
Oil temperature (°C)	[TS]	70.000

4 Geometry

4.1 Reference profiles

Reference profile Gear 1

Reference profile, own input		
Dedendum coefficient	[hfP*]	1.789
Root radius factor	[ρfP*]	0.236
	[ρfPmax*]	0.281
Addendum coefficient	[haP*]	1.352
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Manufacturing center distance (mm)	[a0]	32.776
(mm)	[a0.e/i]	32.668 /32.607
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.710 / 0.719

Reference profile Gear 2

Reference profile, own input		
Dedendum coefficient	[hfP*]	1.509
Root radius factor	[ρfP*]	0.361
	[ρfPmax*]	0.406
Addendum coefficient	[haP*]	1.613
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000

	not topping	
Manufacturing center distance (mm)	[a0]	75.261
(mm)	[a0.e/i]	75.115 /75.038
Smallest radius of curvature, root rounding (mm)	[ρmin.e/i]	0.847 / 0.852

4.1.1 Information on final machining

		Gear 1	Gear 2
Dedendum reference profile	[hfP*]	1.789	1.509
Tooth root radius reference profile	[ρfP*]	0.236	0.361
Addendum reference profile	[haP*]	1.352	1.613
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000

Type of profile modification:	for uniform mesh		
Tip relief (µm)	[Ca L/R]	13.5 /13.5	14.7 /14.7
Root relief (µm)	[Cf L/R]	17.0 /17.0	17.0 /17.0

4.2 Basic data

Overall transmission ratio	[itot]	-2.346
Gear ratio	[u]	2.346
Transverse module (mm)	[mt]	2.458
Transverse pressure angle (°)	[αt]	20.380
Working pressure angle (°)	[αwt]	21.854
(°)	[αwt.e/i]	21.877 /21.831
Working pressure angle at normal section (°)	[αwn]	19.288
Base helix angle (°)	[βb]	27.457
Helix angle at operating pitch circle (°)	[βw]	29.242
Reference center distance (mm)	[ad]	106.932
Sum of profile shift coefficients	[Σxi]	0.5140

		Gear 1	Gear 2
Profile shift coefficient	[x]	0.3810	0.1330
Generating profile shift coefficient	[xE.e/i]	0.3309/ 0.3023	0.0650/ 0.0293
Virtual gear no. of teeth	[zn]	37.753	88.575
Involute length (mm)	[l_dFa-l_dFf]	6.075	6.346

4.3 Diameters and their allowances

		Gear 1	Gear 2
Reference diameter (mm)	[d]	63.913	149.951
Base diameter (mm)	[db]	59.913	140.565
Tip alteration (mm)	[k*mn]	0.000	0.000
Tip diameter (mm)	[da]	71.365	157.460
(mm)	[da.e/i]	71.365 /71.335	157.460 /157.420
Tip diameter allowances (mm)	[Ada.e/i]	0.000 / -0.030	0.000 / -0.040
Tip form diameter (mm)	[dFa]	71.365	157.460
(mm)	[dFa.e/i]	71.365 /71.335	157.460 /157.420
Active tip diameter (mm)	[dNa]	71.365	157.460
(mm)	[dNa.e/i]	71.365 /71.335	157.460 /157.420
V-Circle diameter (mm)	[dv]	65.552	150.523
(mm)	[dv.e/i]	65.336 /65.213	150.231 /150.077

Operating pitch diameter (mm)	[dw]	64.552	151.448
(mm)	[dw.e/i]	64.562 /64.541	151.473 /151.424
Root diameter (mm)	[df]	57.861	144.033
(mm)	[df.e/i]	57.645 /57.522	143.741 /143.587
Active root diameter (mm)	[dNf]	60.653	146.600
(mm)	[dNf.e/i]	60.681 /60.638	146.642 /146.573
Root form diameter (mm)	[dFf]	60.308	145.691
(mm)	[dFf.e/i]	60.240 /60.204	145.472 /145.359

4.4 Tip clearances and tooth heights

		------- Gear 1 ----------	Gear 2 -------
Theoretical tip clearance (mm)	[c]	0.301	0.340
Effective tip clearance (mm)	[c.e/i]	0.556 / 0.430	0.546 / 0.430
Reserve (dNf-dFf)/2 (mm)	[cF.e/i]	0.238 / 0.199	0.641 / 0.550
Addendum, $m_n(h_{aP}$*+x+k) (mm)	[ha]	3.726	3.755
(mm)	[ha.e/i]	3.726 / 3.711	3.755 / 3.735
Dedendum, $m_n(h_{fP}$*-x) (mm)	[hf]	3.026	2.959
(mm)	[hf.e/i]	3.134 / 3.196	3.105 / 3.182
Tooth height (mm)	[h]	6.752	6.713

4.5 Roll angle

		------- Gear 1 ----------	Gear 2 -------
Roll angle at dFa (°)	[ξFa.e/i]	37.080 /37.027	28.924 /28.888
Roll angle to dNf (°)	[ξNf.e/i]	9.207 / 8.942	17.030 /16.931
Roll angle at dFf (°)	[ξFf.e/i]	5.998 / 5.660	15.272 /15.092

4.6 Tooth thickness and pitch

		------- Gear 1 ----------	Gear 2 -------
Tooth thickness on reference circle, arc, in module	[sn*]	1.8184	1.6573
Normal tooth thickness at tip circle (mm)	[san]	0.786	0.768
(mm)	[san.e/i]	0.726 / 0.666	0.687 / 0.618
Normal tooth thickness at tip form circle (mm)	[sFan]	0.786	0.768
(mm)	[sFan.e/i]	0.726 / 0.666	0.687 / 0.618
Normal space width at root circle (mm)	[efn]	0.000	1.544
(mm)	[efn.e/i]	0.000 / 0.000	1.577 / 1.596
Pitch on reference circle (mm)	[pt]	7.723	7.723
Base pitch (mm)	[pbt]	7.239	7.239
Transverse pitch on contact-path (mm)	[pet]	7.239	7.239
Lead height (mm)	[pz]	362.235	849.859
Axial pitch (mm)	[px]	13.932	13.932

4.7 Sliding

		------- Gear 1 ----------	Gear 2 -------
Max. sliding velocity at tip (m/s)	[vga]	7.708	7.624
Specific sliding at the tip	[ζa]	0.542	0.688

Specific sliding at the root	[ζf]	-2.202	-1.185
Mean specific sliding	[ζm]	0.615	
Sliding factor on tip	[Kga]	0.326	0.322
Sliding factor on root	[Kgf]	-0.322	-0.326

4.8 Contact ratios

		Pair
Minimal length of contact line (mm)	[Lmin]	66.170
Transverse contact ratio	[εα]	2.026
	[εα.e/m/i]	2.032 / 2.021/ 2.009
Overlap ratio	[εβ]	2.082
Total contact ratio	[εγ]	4.107
	[εγ.e/m/i]	4.114 / 4.102/ 4.091
Length of path of contact (mm)	[gα]	14.665
(mm)	[gα.e/m/i]	14.712 /14.629/14.546
Approach path of contact (mm)	[gf]	7.292
(mm)	[gf.e/m/i]	7.325 / 7.270/ 7.215
Recess path of contact (mm)	[ga]	7.373
(mm)	[ga.e/m/i]	7.387 / 7.359/ 7.331

		Gear 1	Gear 2
Addendum contact ratio	[ε]	1.018	1.007
	[ε.e/i]	1.020 / 1.013	1.012 / 0.997
Length T1-A and T2-A (mm)	[T1A,T2A]	4.722	35.480
(mm)	[.e/i]	4.675 / 4.814	35.480 / 35.436
Length T1-B and T2-B (mm)	[T1B,T2B]	4.908	35.294
(mm)	[.e/i]	4.908 / 4.881	35.247 / 35.368
Length T1-C and T2-C (mm)	[T1C,T2C]	12.014	28.188
(mm)	[.e/i]	12.000 / 12.028	28.155 / 28.221
Length T1-D and T2-D (mm)	[T1D,T2D]	19.201	21.001
(mm)	[.e/i]	19.154 / 19.292	21.001 / 20.957
Length T1-E and T2-E (mm)	[T1E,T2E]	19.387	20.815
(mm)	[.e/i]	19.387 / 19.359	20.768 / 20.890
Length T1-T2 (mm)	[T1T2]	40.202	
(mm)	[.e/i]	40.155 / 40.249	
Diameter of single contact point B (mm)	[d-B]	60.712	157.293
(mm)	[d-B.e/i]	60.712 / 60.703	157.250 /157.360
Diameter of single contact point D (mm)	[d-D]	71.163	146.706
(mm)	[d-D.e/i]	71.113 / 71.262	146.706 /146.681

5 General influence factors

5.1 Forces and circumferential speed

		Gear 1	Gear 2
Nominal circum. force at pitch circle (N)	[Ft]	10013.54	
Axial force (N)	[Fa]	5550.59	5550.59
Radial force (N)	[Fr]	3720.01	3720.01
Normal force (N)	[Fnorm]	12038.21	12038.21
Nominal circumferential force per mm (N/mm)	[w]	345.29	
Only as information: Forces at operating pitch circle:			
Nominal circumferential force (N)	[Ftw]	9914.53	

Axial force (N)	[Faw]	5550.59	5550.59
Radial force (N)	[Frw]	3976.36	
Circumferential speed reference circle (m/s)	[v]	23.43	
Circumferential speed operating pitch circle (m/s)	[v(dw)]	23.66	

5.2 Contact stiffness

Running-in value (µm)	[yp]	0.6
Running-in value (µm)	[yf]	0.6
Correction factor	[CM]	0.800
Gear blank factor	[CR]	1.000
Basic rack factor	[CB]	0.745
Material coefficient	[E/Est]	1.000
Single stiffness (N/mm/µm)	[c']	10.246
Theoretical single stiffness (N/mm/µm)	[c'th]	19.667
Contact stiffness (N/mm/µm)	[cγα]	18.128
Contact stiffness (N/mm/µm)	[cγβ]	15.408
Reduced mass (kg/mm)	[mRed]	0.01054
Resonance speed (min-1)	[nE1]	15234
Resonance ratio (-)	[N]	0.459
Subcritical range		

5.3 Calculation of K factors

Running-in value (µm)	[yα]	0.3

5.4 K factors

Dynamic factor	[Kv]	1.066
Factor $K_{H\beta}$, own input		
Face load factors		
- Flank	[KHβ]	1.287
- Tooth root	[KFβ]	1.217
- Scuffing	[KBβ]	1.287
Transverse load factors		
- Flank	[KHα]	1.000
- Tooth root	[KFα]	1.000
- Scuffing	[KBα]	1.000
Application factor	[KA]	1.000
Mesh load factor	[Kγ]	1.000

6 Calculation of tooth root strength (fracture)

Calculation of Tooth form coefficients according method: B

		------- Gear 1	---------- Gear 2 -------
Calculated with generating profile shift coefficient	[xE.i]	0.3023	0.0293
Tooth form factor	[YF]	0.722	0.789
Stress correction factor	[YS]	2.153	2.086
Load application angle (°)	[αFen]	14.009	17.231
Load distribution influence factor	[fε]	0.623	
Load application diameter, virtual spur gear (mm)	[d_{en}]	80.642	190.692
Load application diameter (mm)	[d_{en}]	63.386	150.206
Bending moment arm (mm)	[hFe]	2.106	2.374
(-)	[hFe*]	0.980	1.104
Tooth root thickness at critical cross section (mm)	[sFn]	4.894	4.930
(-)	[sFn*]	2.276	2.293

Root fillet radius at critical cross section (mm)		[ρF]	1.109	1.095
	(-)	[ρF*]	0.516	0.509
Tangent contact point, virtual spur gear (mm)		[x,y]	2.447 /29.047	
Tangent contact point, virtual spur gear (mm)		[x,y]		2.465 /72.197
Diameter at critical cross section (mm)		[d_{sFn}]	58.300	144.477
Tangent at critical cross section (°)		[α_{sFn}]	30.000	30.000
Notch parameter		[q_s]	2.207	2.251
Helix angle factor		[Yβ]	1.133	
Deep tooth factor		[YDT]	1.000	
Gear rim thickness (mm)		[sr]	16.26	7.53
Gear rim factor		[YB]	1.00	1.11
Effective facewidth (mm)		[beff]	31.00	29.00
Nominal stress at tooth root (N/mm²)		[σF0]	264.52	332.36
Tooth root stress (N/mm²)		[σF]	343.10	431.09
Permissible bending stress at root of Test-gear				
Notch sensitivity factor		[YδrelT]	0.997	0.998
Surface factor		[YRrelT]	0.957	0.957
Size factor, tooth root		[YX]	1.000	1.000
Life factor		[YNT]	0.866	0.881
$Y_{\delta relT} \cdot Y_{RrelT} \cdot Y_X \cdot Y_{NT}$			0.826	0.841
Alternating bending factor, mean stress influence coefficient				
		[YM]	1.000	1.000
Stress correction factor		[YST]	2.00	
$Y_{ST} \cdot \sigma_{Flim}$ (N/mm²)		[σFE]	850.00	850.00
Permissible tooth root stress σFG/SFmin (N/mm²)		[σFP]	501.49	510.36
Limit strength tooth root (N/mm²)		[σFG]	702.09	714.50

6.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SFmin]	1.40	1.40
Safety for tooth root stress	[SF=σFG/σF]	2.05	1.66
Transmittable power (kW)	[kWRating]	342.86	277.71

7 Calculation of flank strength (pitting)

		------- Gear 1	---------- Gear 2 -------
Zone factor	[ZH]	2.244	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Contact ratio factor	[Zε]	0.703	
Helix angle factor	[Zβ]	1.069	
Effective facewidth (mm)	[beff]	29.00	
Nominal contact stress (N/mm²)	[σH0]	888.27	
Contact stress at operating pitch circle (N/mm²)	[σHw]	1040.40	
Coefficient [fZCa] 1.07 (Helical gear sets with suitable flank modifications acc. to manufacturers experience)			
Single tooth contact factor	[ZB,ZD]	1.03	1.03
Contact stress (N/mm²)	[σHB, σHD]	1076.20	1076.20
Lubricant coefficient for N_L	[ZL]	0.931	0.931
Speed factor at N_L	[ZV]	1.025	1.025
Roughness factor for N_L	[ZR]	0.959	0.959
Material hardening factor for N_L	[ZW]	1.000	1.000
Life factor	[ZNT]	0.874	0.897
Size factor (flank)	[ZX]	1.000	1.000
$Z_L \cdot Z_V \cdot Z_R \cdot Z_{NT} \cdot Z_X$		0.800	0.821
Limited pitting is permissible: No			

Permissible contact stress, σHG/SHmin (N/mm²)	[σHP]	1200.30	1232.11
Pitting stress limit (N/mm²)	[σHG]	1200.30	1232.11

7.1 Safety factors

		------- Gear 1	---------- Gear 2 -------
Required safety	[SHmin]	1.00	1.00
Safety factor for contact stress on operating pitch circle	[SHw]	1.15	1.18
Safety against pressure, σHG/σHBD Single contact	[SHBD]	1.12	1.14
Safety regarding transmittable torque	[SHBD²]	1.24	1.31
Transmittable power (kW)	[kWRating]	291.79	307.46

8 Micropitting

Calculation method according to ISO/TS 6336-22:2018

Calculation has not been carried out, lubricant: Load stage micropitting test not known

9 Scuffing load capacity

Calculation method according to ISO/TS 6336-20/21:2022

		------- Gear 1	---------- Gear 2 -------
Thermal contact coefficient (N/mm/s^.5/K)	[BM]	13.780	13.780
with C_{eq} according to	ISO/TS 6336-20, Annex B		
Optimal tip relief (µm)	[Ceff]	19.05	
C_a taken as optimal in the calculation. 0=no, 1=yes		1	1

		---------- Pair ----------	
Helical load factor for scuffing	[KBγ]	1.300	
Lubrication coefficient for lubrication type	[XS]	1.000	
Scuffing test, load stage	[FZGtest]	FZG - Test A / 8.3 / 90 (ISO 14635 - 1),	12
Multiple meshing factor	[Xmp]	1.000	
Relative structural factor, scuffing	[XWrelT]	1.000	
Effective facewidth (mm)	[beff]	29.000	
Applicable circumferential force/facewidth (N/mm)	[wBt]	473.693	
$K_{B\gamma}$ = 1.300, $w_{Bt} \cdot K_{B\gamma}$ = 615.801			
Angle factor	[Xαβ]	0.976	
ε_1 =1.018, ε_2 =1.007			

9.1 Flash temperature-criteria

		---------- Pair ----------	
Lubricant factor	[XL]	0.881	
Tooth mass temperature (°C)	[θMi]	91.59	
$\theta_{Mi} = \theta_{oil} + X_S \cdot 0.47 \cdot X_{mp} \cdot \theta_{flm}$			
Average flash temperature (Δ°C)	[θflm]	45.93	
Scuffing temperature (°C)	[θS]	365.60	
Γ coordinates (point of highest temperature)	[Γ]	-0.595	
[Γ.A]= -0.607, [Γ.E]= 0.614			
Maximum flash temperature (Δ°C)	[θflamax]	103.18	
Maximum contact temperature (°C)	[θBmax]	194.77	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Approach factor	[XJ]	1.000	
Load sharing factor	[XΓ]	0.661	
Dynamic viscosity (mPa*s)	[ηM]	12.46	(70.0 °C)
Coefficient of friction	[μ_m]	0.052	
Required safety	[SBmin]	2.000	
Margin of safety for scuffing, flash temperature	[SB]	2.369	

9.2 Integral temperature-criteria

		Pair	
Lubricant factor	[XL]	1.000	
Tooth mass temperature (°C)	[θMC]	81.82	
$\theta_{MC} = \theta_{oil} + X_S \cdot 0.70 \cdot \theta_{flaint}$			
Integral scuffing temperature (°C)	[θSint]	365.79	
Flash factor (°K·N^-.75·s^.5·m^-.5·mm)	[XM]	50.058	
Running-in factor, well run in	[XE]	1.000	
Contact ratio factor	[Xε]	0.219	
Dynamic viscosity (mPa*s)	[ηOil]	12.46	(70.0 °C)
Mean coefficient of friction	[μ_m]	0.057	
Geometry factor	[XBE]	0.296	
Meshing factor	[XQ]	1.000	
Tip relief factor	[XCa]	1.981	
Mean flash temperature (Δ°C)	[θflaint]	16.89	
Integral tooth flank temperature (°C)	[θint]	107.16	
Required safety	[SSmin]	1.800	
Safety factor for scuffing (intg.-temp.)	[SSint]	3.414	
Safety factor for transmitted moment (int.-T.)	[SSL]	7.960	

10 Measurements for tooth thickness

10.1 Tooth thickness tolerances

		Gear 1	Gear 2
Tooth thickness tolerance		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness allowance (normal section) (mm)	[Asn.e/i]	-0.070 / -0.110	-0.095 / -0.145

10.2 Base tangent lengths

		Gear 1	Gear 2
Number of teeth spanned	[k]	5.000	10.000
Base tangent length (no backlash) (mm)	[Wk]	30.254	63.174
Base tangent length with allowance (mm)	[Wk.e/i]	30.187 /30.149	63.084 /63.036
(mm)	[ΔWk.e/i]	-0.067 / -0.105	-0.090 / -0.138
Diameter of measuring circle (mm)	[dMWk.m]	65.621	151.293

> Gear 2: Base tangent length cannot be measured (gear too thin)

10.3 Measurement over balls and pins

		Gear 1	Gear 2
Theoretical diameter of ball/pin (mm)	[DM]	3.794	3.594
Effective diameter of ball/pin (mm)	[DMeff]	4.250	4.250
Radial single-ball measurement, no backlash (mm)	[MrK]	36.169	78.918
Radial single-ball measurement (mm)	[MrK.e/i]	36.090 /36.044	78.791 /78.724
Diameter of measuring circle (mm)	[dMMr.m]	66.194	151.802
Diametral measurement over two balls, no backlash (mm)	[MdK]	72.338	157.785
Diametral measurement over two balls (mm)	[MdK.e/i]	72.179 /72.088	157.532 /157.397
Diametral measurement over pins, no backlash (mm)	[MdR]	72.338	157.836
Measurement over pins according to DIN 3960 (mm)	[MdR.e/i]	72.179 /72.088	157.583 /157.448
Measurement over 2 pins, free, AGMA 2002 (mm)	[dk2f.e/i]	0.000 / 0.000	157.508 /157.373
Measurement over 2 pins, transverse, AGMA 2002 (mm)	[dk2t.e/i]	0.000 / 0.000	157.633 /157.498
Measurement over 3 pins, axial, AGMA 2002 (mm)	[dk3A.e/i]	72.179 /72.088	157.583 /157.448

10.4 Tooth thickness

		Gear 1	Gear 2
Medium tip diameter (mm)	[da.m]	71.350	157.440

Reference chordal height from da.m (mm)	[hac]	3.764	3.761
Tooth thickness at height hac, chord (mm)	[sc]	3.908	3.563
(mm)	[sc.e/i]	3.839 / 3.800	3.468 / 3.419
Tooth thickness on reference circle, arc (mm)	[sn]	3.910	3.563
(mm)	[sn.e/i]	3.840 / 3.800	3.468 / 3.418

10.5 Backlash

		Pair
Circumferential backlash from Aa (mm)	[jtw_Aa.e/i]	0.014 / -0.014
Radial backlash (mm)	[jrw.e/i]	0.389 / 0.222
Circumferential backlash (transverse section) (mm)	[jtw.e/i]	0.309 / 0.177
Normal backlash (mm)	[jnw.e/i]	0.253 / 0.146
Center distance allowances (mm)	[Aa.e/i]	0.018 / -0.018
Backlash free center distance (mm)	[aControl.e/i]	107.761 /107.629
Backlash free center distance, allowances (mm)	[jta]	-0.239 / -0.371

		Gear 1	Gear 2
dNf.i with aControl (mm)	[dNf0.i]	60.372	146.040
Reserve (dNf0.i-dFf.e)/2 (mm)	[cF0.i]	0.066	0.284
Tip clearance (mm)	[c0.i(aControl)]	0.076	0.076

Torsional angle Gear 2 with fixed Gear 1:			
Total torsional angle (°)	[j.tSys]	0.2334/ 0.1335	(0°14'0"/ 0°8'1")
Torsional angle Gear 1 with fixed Gear 2:			
Total torsional angle (°)	[j.tSys]	0.5477/ 0.3133	(0°32'52"/ 0°18'48")

11 Gear tolerances

		Gear 1	Gear 2
According to ISO 1328-1:2013, ISO 1328-2:1997			
Accuracy grade	[Q]	A6	A6
Single pitch tolerance (µm)	[fpT]	±8.50	±8.50
Base pitch tolerance (µm)	[fpbT]	±7.97	±7.97
Sector pitch tolerance (µm)	[Fpz/8T]	±16.00	±19.00
Cumulative pitch tolerance, total (µm)	[FpT]	25.00	29.00
Adjacent pitch difference tolerance (µm)	[fuT]	12.00	12.00
Profile form tolerance (µm)	[ffαT]	8.50	8.50
Profile slope tolerance (µm)	[fHαT]	±7.00	±7.00
Profile tolerance, total (µm)	[FαT]	11.00	11.00
Helix form tolerance (µm)	[ffβT]	10.00	10.00
Helix slope tolerance (µm)	[fHβT]	±9.00	±9.00
Helix tolerance, total (µm)	[FβT]	13.00	14.00
Runout tolerance (µm)	[FrT]	23.00	26.00
Single flank composite tolerance, total (µm)	[FisT]	34.00	37.00
Single flank composite tolerance, tooth-to-tooth (µm)	[fisT]	8.00	8.00
Radial composite tolerance, total (µm)	[Fi"]	31.00	37.00
Radial composite tolerance, tooth-to-tooth (µm)	[fi"]	9.50	9.50

F_{idT} (F_i"), f_{idT} (f_i") according to ISO 1328-2:1997 calculated with the geometric mean values for m_n and d.

According to ISO 1328-2:2020			
Accuracy grade	[R]	R39	R39
Radial composite tolerance, total (µm)	[FidT]	29.00	32.00
Radial composite tolerance, tooth-to-tooth (µm)	[fidT]	13.00	13.00

Axis alignment tolerances recommendation according to ISO/TR 10064-3:1996, AGMA 915-3-A99		
Total helix deviation for calculation (µm)	[Fβ]	14.00

Distance between bearings for calculation (mm)	[LG]	62.00
Maximum value for axis deviation error (µm)	[fΣβ]	± 14.97
Maximum value for axis inclination error (µm)	[fΣδ]	± 29.93

12 Modifications and determination of the tooth form

12.1 Profile and flank line modifications

Gear 1

Symmetric (both flanks)

- Tip relief, arc-like

  Caa = 12.000 µm LCa = 1.260 ·mn dCa = 68.572 mm

- Shortened profile crowning

  Caf = 17.0 µm F1 = 0.500 F2 = 0.250

  rcrown=3224mm, Cαa = 1.475µm, dCa = 68.584mm, Factor1 =1.25452 (combine with tip relief)

- Flank line crowning

  Cb = 5.000 µm

  bx=31.000mm, rcrown=24025mm

Gear 2

Symmetric (both flanks)

- Tip relief, arc-like

  Caa = 12.000 µm LCa = 1.619 ·mn dCa = 154.448 mm

- Flank line crowning

  Cb = 5.000 µm

  bx=29.000mm, rcrown=21025mm

- Shortened profile crowning

  Caf = 17.0 µm F1 = 0.500 F2 = 0.250

  rcrown=2642mm, Cαa = 2.741µm, dCa = 154.463mm, Factor1 =1.61042 (combine with tip relief)

12.1.1 Tip relief verification

		------- Gear 1	---------- Gear 2 -------
Diameter (mm)	[dcheck]	71.292	157.377
Tip relief left/right (µm)	[Ca L/R]	13.0 /13.0	14.3 /14.3
Diameter (mm)	[dNa.i]	71.335	157.420
Tip relief left/right (µm)	[Ca L/R]	13.4 /13.4	14.6 /14.6

12.2 Data for the tooth form calculation

Data not available.
Please run the calculation in the "Tooth form" tab and open the main report again.

13 Supplementary data

Maximal possible centre distance (ε_α = 1.0)	[aMAX]	110.978

13.1 Masses, stiffnesses and moments of inertia

		------- Gear 1	---------- Gear 2 -------
Mass (kg)	[m]	0.677	1.106
Total mass (kg)	[Σm]	1.783	
Moment of inertia for system, relative to the input:			
calculation without consideration of the exact tooth shape			
Gears individually ((da+df)/2...di) (kg*m²)	[J]	0.000406	0.005427
System (da+df)/2...di (kg*m²)	[J]	0.001392	

13.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:

Torsional stiffness (MNm/rad)	[cr]	0.397
Torsion when subjected to nominal torque (°)	[δcr]	0.046
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	82.138

13.3 Power loss

Average coefficient of friction according to Niemann	[μ_m]	0.051
Wear sliding coefficient by Niemann	[ζw]	1.245
Loss factor	[HV]	0.197
Gear power loss (kW)	[PVZ]	2.369
Meshing efficiency (%)	[ηz]	98.990

14 Service life, damage

Required safety for tooth root	[S_{Fmin}]	1.4000
Required safety for tooth flank	[S_{Hmin}]	1.0000
Required service life	[H]	9500.0000

Service life (calculated with required safeties):

System service life (h)	[H_{att}]	>	1000000

		------- Gear 1	---------- Gear 2 -------
Tooth root service life (h)	[H_{Fatt}]	1e+06	1e+06
Tooth flank service life (h)	[H_{Hatt}]	1e+06	1e+06

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

14.1 Damage

Damage relative to the required service life (H, 9500.0 h)

D_{F1} (%)	<TD CENTER>D_{F2} (%)	<TD CENTER>D_{H1} (%)	<TD CENTER>D_{H2} (%)
0.0000	0.0000	0.0000	0.0000

15 Reliability calculation

15.1 Calculation method

Calculation method according to B. Bertsche, Reliability in Automotive and Mechanical Engineering, Springer-Verlag Berlin Heidelberg 2008

15.2 Factors

Reliability of material data for σ_{Hlim} (%)	[$R\sigma_{Hlim}$]	99.00
Reliability of material data for σ_{Flim} (%)	[$R\sigma_{Flim}$]	99.00

Calculation of coefficients for reliability R(t)
$R(t) = 100 \cdot Exp(-((t \cdot fac - t_0)/(T - t_0))^{\beta})$ (%)

Gear	Type	fac	<TD CENTER>β	<TD CENTER>t_0	<TD CENTER>T	<TD CENTER>R(H)
		cycles/h		cycles	cycles	%
1	Tooth root	420000	1.700e+00	9.654e+29	1.484e+30	100.0000
1	Tooth flank	420000	1.300e+00	9.014e+29	4.295e+30	100.0000
2	Tooth root	179016	1.700e+00	9.654e+29	1.484e+30	100.0000
2	Tooth flank	179016	1.300e+00	9.014e+29	4.295e+30	100.0000

fac = Number of load cycles per hour
β = Weibull shape parameter
t_0 = Failure-free number of load cycles
T = Characteristic service life (in load cycles) for 63.2% failure probability
R(H) = Reliability for required service life

15.3 Resulting reliabilities and service lifes

Required service life (h)	[H_{min}]	9500.0000
Reliability R, tooth roots subsystem (%)	[R_{subF}]	100.0000
Reliability R, tooth flanks subsystem (%)	[R_{subH}]	100.0000
Reliability R, gears subsystem (%)	[R_{subG}]	100.0000
Required reliability (%)	[R_{min}]	99.0000
Service life H, tooth roots subsystem (h)	[H_{subF}]	> 1'000'000
Service life H, tooth flanks subsystem (h)	[H_{subH}]	> 1'000'000
Service life H, gears subsystem (h)	[H_{subG}]	> 1'000'000

16 Remarks

16.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.

- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.

16.2 Calculations and factors

- Calculation of Z_β acc. to Corrigendum 1 ISO 6336-2:2008 with Z_β = 1/(COS(β)^0.5)
- Details of calculation method:
  c_γ according to Method B
  K_v according to Method B
- The logarithmically interpolated value must be entered for factors Z_L, Z_V, Z_R, Z_W, Z_X, $Y_{\delta relT}$, Y_{RrelT} and Y_X. This value is calculated from the infinite life strength and static strength values, based on the number of load cycles.

End of report (lines: 654)
